

```

#include <WiFi.h>
#include <WebServer.h>

/* * SMART LIBRARY ESP32 - 20 BOOKS EDITION
* * Features: Access Point, Offline Mode, 20 Books (6 Avail, 14 Checked Out)
* *
* * WIRING INSTRUCTIONS (6 LEDs):
* * Connect LEDs to these 6 pins ONLY:
* * Row 1: Pin 13
* * Row 2: Pin 12
* * Row 3: Pin 14
* * Row 4: Pin 27
* * Row 5: Pin 26
* * Row 6: Pin 25
*/

// ===== CONFIGURATION =====
const char* ap_ssid = "Smart Library";
const char* ap_password = "12345678";

// THE 6 AVAILABLE PINS
const int libraryPins[] = {13, 12, 14, 27, 26, 25};
const int pinCount = sizeof(libraryPins) / sizeof(libraryPins[0]);

// Timers
unsigned long ledOnStartTime = 0;
int activeLedPin = 0;
const long ledTimeout = 5000;

WebServer server(80);

// ===== HTML/CSS/JS =====
const char index_html[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Smart Library</title>
  <style>
    /* --- STYLES --- */
    :root { --primary: #4f46e5; --bg: #f8fafc; --text: #1e293b; --white: #ffffff; }
    * { box-sizing: border-box; }
    html, body { height: 100%; margin: 0; padding: 0; font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
    Roboto, sans-serif; background: var(--bg); color: var(--text); }

    /* Clock */
    .datetime-display { position: absolute; top: 20px; right: 25px; text-align: right; z-index: 150; line-height: 1.2; }
    .time-text { font-size: 1.5rem; font-weight: 700; display: block; }
    .date-text { font-size: 0.85rem; opacity: 0.8; font-weight: 500; text-transform: uppercase; }
    #home-view .datetime-display { color: rgba(255,255,255,0.9); }
    #app-view .datetime-display { color: var(--primary); top: 18px; }

    /* Home Screen */

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    #home-view { position: fixed; top: 0; left: 0; width: 100%; height: 100%; background: linear-gradient(135deg,
    #1e1b4b 0%, #312e81 100%); color: white; display: flex; flex-direction: column; justify-content: center; align-items:
    center; z-index: 100; text-align: center; padding: 20px; transition: opacity 0.5s ease; }
    .home-title { font-size: 3rem; margin: 0; font-weight: 800; }
    .home-subtitle { color: #c7d2fe; font-size: 1.2rem; margin-top: 10px; margin-bottom: 40px; }
    .enter-btn { background: white; color: #312e81; border: none; padding: 16px 40px; font-size: 1.2rem; font-weight:
    700; border-radius: 50px; cursor: pointer; box-shadow: 0 10px 25px rgba(0,0,0,0.2); transition: transform 0.2s;
    display: flex; align-items: center; gap: 10px; }
    .enter-btn:hover { transform: translateY(-3px); }

    /* App Screen */
    #app-view { display: none; min-height: 100vh; padding-bottom: 80px; }
    .navbar { background: var(--white); padding: 1rem 1.5rem; box-shadow: 0 1px 3px rgba(0,0,0,0.1); display: flex;
    justify-content: space-between; align-items: center; position: sticky; top: 0; z-index: 10; height: 70px; }
    .logo { font-weight: 800; font-size: 1.25rem; color: var(--primary); display: flex; align-items: center; gap: 8px;
    cursor: pointer; }

    .container { max-width: 1000px; margin: 0 auto; padding: 20px; width: 100%; }
    .search-container { background: white; padding: 20px; border-radius: 16px; box-shadow: 0 4px 6px -1px rgba(0,
    0, 0, 0.1); margin-bottom: 30px; }
    .search-input { width: 100%; padding: 16px; font-size: 1.1rem; border: 2px solid #e2e8f0; border-radius: 12px;
    outline: none; }
    .search-input:focus { border-color: var(--primary); }

    .grid { display: grid; grid-template-columns: repeat(auto-fill, minmax(300px, 1fr)); gap: 20px; }

    .card { background: var(--white); border-radius: 12px; padding: 20px; box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    border-left: 5px solid var(--primary); display: flex; flex-direction: column; justify-content: space-between; position:
    relative; overflow: hidden; }
    .card.checked-out { border-left-color: #ef4444; background: #fff1f2; }

    .badge-row { display: flex; justify-content: space-between; margin-bottom: 10px; }
    .badge { padding: 4px 8px; border-radius: 4px; font-size: 0.75rem; font-weight: 700; text-transform: uppercase; }
    .badge-cat { background: #eef2ff; color: var(--primary); }
    .badge-avail { background: #dcfce7; color: #15803d; }
    .badge-out { background: #fee2e2; color: #b91c1c; }

    .book-title { font-size: 1.25rem; font-weight: 700; margin: 0 0 5px 0; color: #0f172a; }
    .book-author { color: #64748b; margin: 0 0 15px 0; }
    .location-bar { background: #f1f5f9; padding: 8px 12px; border-radius: 8px; display: flex; justify-content:
    space-between; align-items: center; margin-bottom: 15px; font-family: monospace; color: #475569; font-size: 0.9rem;
    font-weight: 600; }
    .pin-circle { background: #cbd5e1; color: #334155; padding: 2px 8px; border-radius: 6px; }

    /* Buttons */
    .btn-group { display: flex; gap: 10px; }
    .btn { flex: 1; padding: 12px; border: none; border-radius: 8px; font-weight: 700; cursor: pointer; font-size:
    0.9rem; transition: transform 0.1s; }
    .btn:active { transform: scale(0.95); }
    .btn-primary { background: var(--primary); color: white; flex: 2; }
    .btn-secondary { background: #e2e8f0; color: #475569; flex: 1; }
    .btn-disabled { background: #cbd5e1; color: #94a3b8; cursor: not-allowed; flex: 2; }

    /* MODAL (Popup) STYLES */

```

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.modal-overlay { position: fixed; top: 0; left: 0; width: 100%; height: 100%; background: rgba(0,0,0,0.6); z-index:
200; display: none; align-items: center; justify-content: center; padding: 20px; backdrop-filter: blur(4px); }
.modal-content { background: white; padding: 30px; border-radius: 20px; max-width: 500px; width: 100%;
box-shadow: 0 20px 25px -5px rgba(0, 0, 0, 0.1); position: relative; animation: slideUp 0.3s ease-out; }
@keyframes slideUp { from { transform: translateY(50px); opacity: 0; } to { transform: translateY(0); opacity: 1; } }

```

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.modal-header { display: flex; justify-content: space-between; align-items: start; margin-bottom: 20px;
border-bottom: 1px solid #e2e8f0; padding-bottom: 15px; }
.modal-title { font-size: 1.5rem; font-weight: 800; color: #1e293b; margin: 0; line-height: 1.2; }
.close-modal { background: none; border: none; font-size: 1.5rem; color: #94a3b8; cursor: pointer; padding: 0; }

```

```

.detail-row { display: flex; margin-bottom: 12px; }
.detail-label { font-weight: 700; color: #64748b; width: 100px; flex-shrink: 0; }
.detail-value { color: #1e293b; }
.desc-box { background: #f8f9fc; padding: 15px; border-radius: 10px; color: #475569; line-height: 1.5;
margin-top: 15px; font-size: 0.95rem; border: 1px solid #e2e8f0; }

```

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/* Toast */

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.toast { position: fixed; bottom: 30px; left: 50%; transform: translateX(-50%) translateY(100px); background:
#333; color: white; padding: 12px 24px; border-radius: 50px; opacity: 0; transition: all 0.3s; z-index: 300; font-weight:
500; white-space: nowrap; }
.toast.show { opacity: 1; transform: translateX(-50%) translateY(0); }
.toast.error { background: #dc2626; }

```

```

@media (max-width: 480px) { .datetime-display { right: 15px; top: 15px; } .time-text { font-size: 1rem; } .date-text
{ font-size: 0.7rem; } }

```

```

</style>

```

```

</head>

```

```

<body>

```

```

<div id="home-view">
  <div class="datetime-display">
    <span class="time-text" id="clock-home-time">--:--</span>
    <span class="date-text" id="clock-home-date">Loading...</span>
  </div>
  <div style="margin-bottom:20px; color:#818cf8;"><svg width="80" height="80" fill="none" stroke="currentColor"
stroke-width="1.5" viewBox="0 0 24 24"><path d="m16 6 4 14M12 6v14M8 8v12M4 16M4 4h16"/></svg></div>
  <h1 class="home-title">Smart Library</h1>
  <p class="home-subtitle">Connect to "Smart Library" WiFi</p>
  <button class="enter-btn" onclick="enterApp()">Enter Library</button>
</div>

```

```

<div id="app-view">
  <div class="datetime-display">
    <span class="time-text" id="clock-app-time">--:--</span>
    <span class="date-text" id="clock-app-date">Loading...</span>
  </div>
  <div class="navbar">
    <div class="logo" onclick="goHome()">SmartLib</div>
  </div>
  <div class="container">
    <div class="search-container">
      <input type="text" id="searchInput" class="search-input" placeholder="Search across 20 books..."
onkeyup="renderBooks()">
    </div>
  </div>

```

```

    </div>
    <div id="bookGrid" class="grid"></div>
  </div>
</div>

<div id="detailModal" class="modal-overlay" onclick="closeModal(event)">
  <div class="modal-content">
    <div class="modal-header">
      <h2 class="modal-title" id="m-title">Book Title</h2>
      <button class="close-modal" onclick="closeModal(null)">&times;</button>
    </div>

    <div class="detail-row">
      <span class="detail-label">Author:</span>
      <span class="detail-value" id="m-author">Author Name</span>
    </div>
    <div class="detail-row">
      <span class="detail-label">Status:</span>
      <span class="detail-value" id="m-status-text">Available</span>
    </div>
    <div class="detail-row">
      <span class="detail-label">Edition:</span>
      <span class="detail-value" id="m-edition">1st Edition</span>
    </div>
    <div class="detail-row">
      <span class="detail-label">Launched:</span>
      <span class="detail-value" id="m-date">Jan 2020</span>
    </div>

    <div class="desc-box" id="m-desc">
      Description goes here.
    </div>

    <div style="margin-top: 25px;">
      <button id="m-locate-btn" class="btn btn-primary" style="width:100%">LOCATE ON SHELF</button>
    </div>
  </div>
</div>

<div id="toast" class="toast">Notification</div>

<script>
  // --- DATABASE: 20 BOOKS (6 Available + 14 Unavailable) ---
  const books = [
    // --- THE 6 AVAILABLE BOOKS (Assigned to Valid Pins) ---
    { status: 'avail', title: "Deep Learning", author: "Ian Goodfellow", category: "AI & ML", row: "Row 1", gpioPin: 13, edition: "1st Edition", date: "Nov 2016", desc: "The Bible of Deep Learning." },
    { status: 'avail', title: "Artificial Intelligence", author: "Russell & Norvig", category: "AI & ML", row: "Row 2", gpioPin: 12, edition: "4th Edition", date: "Apr 2020", desc: "The standard textbook on AI." },
    { status: 'avail', title: "Clean Code", author: "Robert C. Martin", category: "CSE", row: "Row 3", gpioPin: 14, edition: "1st Edition", date: "Aug 2008", desc: "A Handbook of Agile Software Craftsmanship." },
    { status: 'avail', title: "Intro to Algorithms", author: "Thomas H. Cormen", category: "CSE", row: "Row 4", gpioPin: 27, edition: "3rd Edition", date: "Jul 2009", desc: "Comprehensive introduction to algorithms." },
  ]

```

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    { status: 'avail', title: "Python Data Analysis", author: "Wes McKinney", category: "Data Sci", row: "Row 5",
      gpioPin: 26, edition: "3rd Edition", date: "Aug 2022", desc: "Guide to pandas and data analysis." },
    { status: 'avail', title: "Microelectronic Circuits", author: "Sedra & Smith", category: "ECE", row: "Row 6",
      gpioPin: 25, edition: "8th Edition", date: "Nov 2019", desc: "The gold standard in teaching circuits." },

    // --- THE 14 UNAVAILABLE BOOKS (No Pin) ---
    { status: 'out', title: "Signals and Systems", author: "Oppenheim", category: "ECE", row: "Row 6", gpioPin: 0,
      edition: "2nd Edition", date: "Aug 1996", desc: "Exploration of signals and systems concepts. Checked out." },
    { status: 'out', title: "Digital Logic Design", author: "Morris Mano", category: "ECE", row: "Row 6", gpioPin: 0,
      edition: "5th Edition", date: "Jan 2013", desc: "Classic text on digital design. Checked out." },
    { status: 'out', title: "CMOS VLSI Design", author: "Weste & Harris", category: "VLSI", row: "Row 6", gpioPin: 0,
      edition: "4th Edition", date: "Mar 2010", desc: "Modern CMOS VLSI design issues. Missing." },
    { status: 'out', title: "Engineering Physics", author: "H.K. Malik", category: "General", row: "Row 6", gpioPin: 0,
      edition: "2nd Edition", date: "Jan 2017", desc: "Guide to engineering physics concepts. Checked out." },
    { status: 'out', title: "Circuit Analysis", author: "Hayt & Kemmerly", category: "EEE", row: "Row 6", gpioPin: 0,
      edition: "9th Edition", date: "Mar 2018", desc: "Student-friendly approach to circuit analysis. Repair." },
    { status: 'out', title: "Control Systems", author: "Norman Nise", category: "EEE", row: "Row 6", gpioPin: 0,
      edition: "7th Edition", date: "Feb 2015", desc: "Practical application of control systems. Checked out." },
    { status: 'out', title: "Thermodynamics", author: "P.K. Nag", category: "Mech", row: "Row 6", gpioPin: 0, edition:
      "6th Edition", date: "Jul 2017", desc: "Laws of thermodynamics and applications. Missing." },
    { status: 'out', title: "Fluid Mechanics", author: "Frank M. White", category: "Mech", row: "Row 6", gpioPin: 0,
      edition: "8th Edition", date: "Jan 2015", desc: "Introduction to fluid mechanics. Checked out." },
    { status: 'out', title: "Machine Design", author: "V.B. Bhandari", category: "Mech", row: "Row 6", gpioPin: 0,
      edition: "4th Edition", date: "Jan 2016", desc: "Design of machine elements. Checked out." },
    { status: 'out', title: "Structural Analysis", author: "Hibbeler", category: "Civil", row: "Row 6", gpioPin: 0, edition:
      "10th Edition", date: "Jul 2017", desc: "Theory and application of structural analysis. Repair." },
    { status: 'out', title: "Surveying Vol 1", author: "B.C. Punmia", category: "Civil", row: "Row 6", gpioPin: 0,
      edition: "17th Edition", date: "Jan 2016", desc: "Standard topics in surveying. Checked out." },
    { status: 'out', title: "Concrete Technology", author: "M.S. Shetty", category: "Civil", row: "Row 6", gpioPin: 0,
      edition: "7th Edition", date: "Jan 2018", desc: "Concrete materials and mix design. Missing." },
    { status: 'out', title: "Highway Engineering", author: "Khanna & Justo", category: "Civil", row: "Row 6", gpioPin:
      0, edition: "10th Edition", date: "Feb 2015", desc: "Highway engineering planning and design. Checked out." },
    { status: 'out', title: "Intro to Robotics", author: "John Craig", category: "Mech", row: "Row 6", gpioPin: 0,
      edition: "4th Edition", date: "2018", desc: "Mechanics of robotic manipulators. Checked out." }
  ];

  // --- LOGIC ---
  let idleTimer;
  const IDLE_LIMIT = 2 * 60 * 1000;

  function updateClock() {
    const now = new Date();
    const time = now.toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' });
    const date = now.toLocaleDateString([], { weekday: 'short', day: 'numeric', month: 'short', year: 'numeric' });
    document.getElementById('clock-home-time').textContent = time;
    document.getElementById('clock-home-date').textContent = date;
    document.getElementById('clock-app-time').textContent = time;
    document.getElementById('clock-app-date').textContent = date;
  }
  setInterval(updateClock, 1000); updateClock();

  function resetIdleTimer() {
    clearTimeout(idleTimer);
    if (document.getElementById('app-view').style.display === 'block') {

```

```

        idleTimer = setTimeout(goHome, IDLE_LIMIT);
    }
}

function enterApp() {
    document.getElementById('home-view').style.opacity = '0';
    setTimeout(() => {
        document.getElementById('home-view').style.display = 'none';
        document.getElementById('app-view').style.display = 'block';
        resetIdleTimer();
    }, 500);
}

function goHome() {
    clearTimeout(idleTimer);
    document.getElementById('app-view').style.display = 'none';
    document.getElementById('home-view').style.display = 'flex';
    setTimeout(() => document.getElementById('home-view').style.opacity = '1', 50);
    document.getElementById('searchInput').value = '';
    closeModal(null);
    renderBooks();
}

// --- RENDER ---
function renderBooks() {
    const term = document.getElementById('searchInput').value.toLowerCase();
    const grid = document.getElementById('bookGrid');
    grid.innerHTML = '';
    const filtered = books.filter(b => b.title.toLowerCase().includes(term) ||
b.category.toLowerCase().includes(term));

    if(filtered.length === 0) { grid.innerHTML = '<p style="grid-column:1/-1;text-align:center;color:#888;">No books
found.</p>'; return; }

    filtered.forEach((b, index) => {
        const isAvail = b.status === 'avail';
        const card = document.createElement('div');
        card.className = isAvail ? 'card' : 'card checked-out';

        card.innerHTML = `
            <div>
                <div class="badge-row">
                    <span class="badge badge-cat">${b.category}</span>
                    <span class="badge ${isAvail ? 'badge-avail' : 'badge-out'}">${isAvail ? 'AVAIL' : 'OUT'}</span>
                </div>
                <h3 class="book-title">${b.title}</h3>
                <p class="book-author">by ${b.author}</p>
                <div class="location-bar" style="opacity:${isAvail ? 1 : 0.5}">
                    <span>${b.row}</span>
                    <span>${isAvail ? 'Pin: ' + b.gpioPin : 'Unavailable'}</span>
                </div>
            </div>
            <div class="btn-group">
                <button onclick="showDetails(${index})" class="btn btn-secondary">Info</button>

```

```

        <button onclick="locate(${b.gpioPin}, '${b.row}', ${isAvail})" class="btn ${isAvail ? 'btn-primary' :
'btn-disabled'}">${isAvail ? 'Locate' : 'Checked Out'}</button>
    </div>
    `;
    grid.appendChild(card);
});
}

```

```

// --- MODAL LOGIC ---

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```

function showDetails(index) {

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```

    resetIdleTimer();
    const b = books[index];
    const isAvail = b.status === 'avail';

```

```

    document.getElementById('m-title').textContent = b.title;
    document.getElementById('m-author').textContent = b.author;

```

```

    // Status Color in Modal

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```

    const sText = document.getElementById('m-status-text');
    sText.textContent = isAvail ? "Available on Shelf" : "Checked Out";
    sText.style.color = isAvail ? "#15803d" : "#b91c1c";
    sText.style.fontWeight = "bold";

```

```

    document.getElementById('m-edition').textContent = b.edition;
    document.getElementById('m-date').textContent = b.date;
    document.getElementById('m-desc').textContent = b.desc;

```

```

    // Set locate button in modal

```

```

    const btn = document.getElementById('m-locate-btn');
    if(isAvail) {
        btn.className = "btn btn-primary";
        btn.textContent = "LOCATE ON SHELF";
        btn.onclick = function() { closeModal(null); locate(b.gpioPin, b.row, true); };
    } else {
        btn.className = "btn btn-disabled";
        btn.textContent = "CURRENTLY UNAVAILABLE";
        btn.onclick = null;
    }
}

```

```

    const modal = document.getElementById('detailModal');
    modal.style.display = 'flex';
}

```

```

function closeModal(e) {
    if (e === null || e.target.id === 'detailModal') {
        document.getElementById('detailModal').style.display = 'none';
    }
}

```

```

function locate(pin, row, isAvail) {

```

```

    resetIdleTimer();
    if(!isAvail) return;

```

```

    const t = document.getElementById('toast');

```

```

        t.textContent = `Go to ${row}`;
        t.className = "toast show";
        setTimeout(() => t.className = "toast", 4000);
        fetch(`/locate?pin=${pin}`).catch(e => console.log(e));
    }

    ['mousemove', 'mousedown', 'keypress', 'touchstart', 'scroll'].forEach(e => document.addEventListener(e,
resetIdleTimer));
    renderBooks();
</script>
</body>
</html>
)rawliteral";

// ===== BACKEND =====
void handleRoot() { server.send(200, "text/html", index_html); }

void handleLocate() {
    if (server.hasArg("pin")) {
        int targetPin = server.arg("pin").toInt();
        bool valid = false;
        for(int p : libraryPins) { if(p == targetPin) valid = true; }

        if(valid) {
            for(int p : libraryPins) digitalWrite(p, LOW);
            digitalWrite(targetPin, HIGH);
            activeLedPin = targetPin;
            ledOnStartTime = millis();
            server.send(200, "text/plain", "OK");
        } else {
            server.send(400, "text/plain", "Invalid Pin");
        }
    } else { server.send(400, "text/plain", "Missing Pin"); }
}

void setup() {
    Serial.begin(115200);
    for (int p : libraryPins) { pinMode(p, OUTPUT); digitalWrite(p, LOW); }

    Serial.println("Setting up Access Point...");
    WiFi.softAP(ap_ssid, ap_password);
    Serial.print("AP Created! Connect to: "); Serial.println(ap_ssid);
    Serial.print("Go to IP: "); Serial.println(WiFi.softAPIP());
    server.on("/", handleRoot);
    server.on("/locate", handleLocate);
    server.begin();
}

void loop() {
    server.handleClient();
    if (activeLedPin != 0 && (millis() - ledOnStartTime >= ledTimeout)) {
        digitalWrite(activeLedPin, LOW);
        activeLedPin = 0;
    }
}

```